THE ADVANSWALLET SOLUTION FOR BANKS

Technical Document for AdvansWallet

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AdvansWallet Technical Document

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Introduction: AdvansWallet!

The **AdvansWallet** is a solution that allows individuals to create a wallet on the go, fund it and start transactions.

Key Features

Comes with an Android and IOS App that allows:

- Cash Deposit
- Cash Withdrawal
- Bills Payment
- Airtime Top-up
- Agent Banking App/Portal (for Agents & SuperAgents)
- Dashboard & Reports
- System Configurations
- User Management (Maker–Checker rules are enforced)
- Fees Configurations
- And many more

It also comes with standard API to integrate with Banks core banking system system

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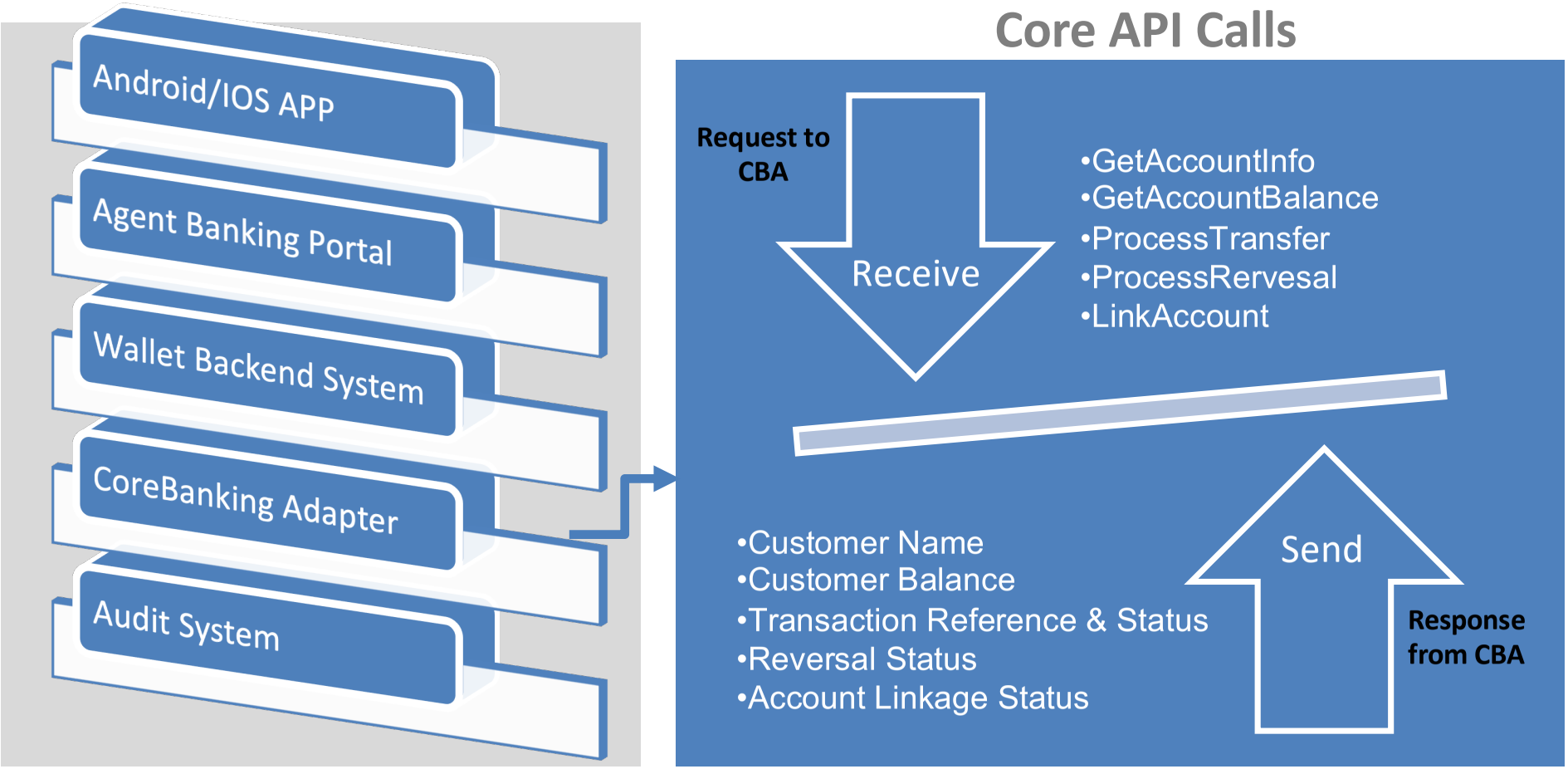
Modular Components

Our proposed solution has 5 key services that will address core touchpoints of the business requirement.



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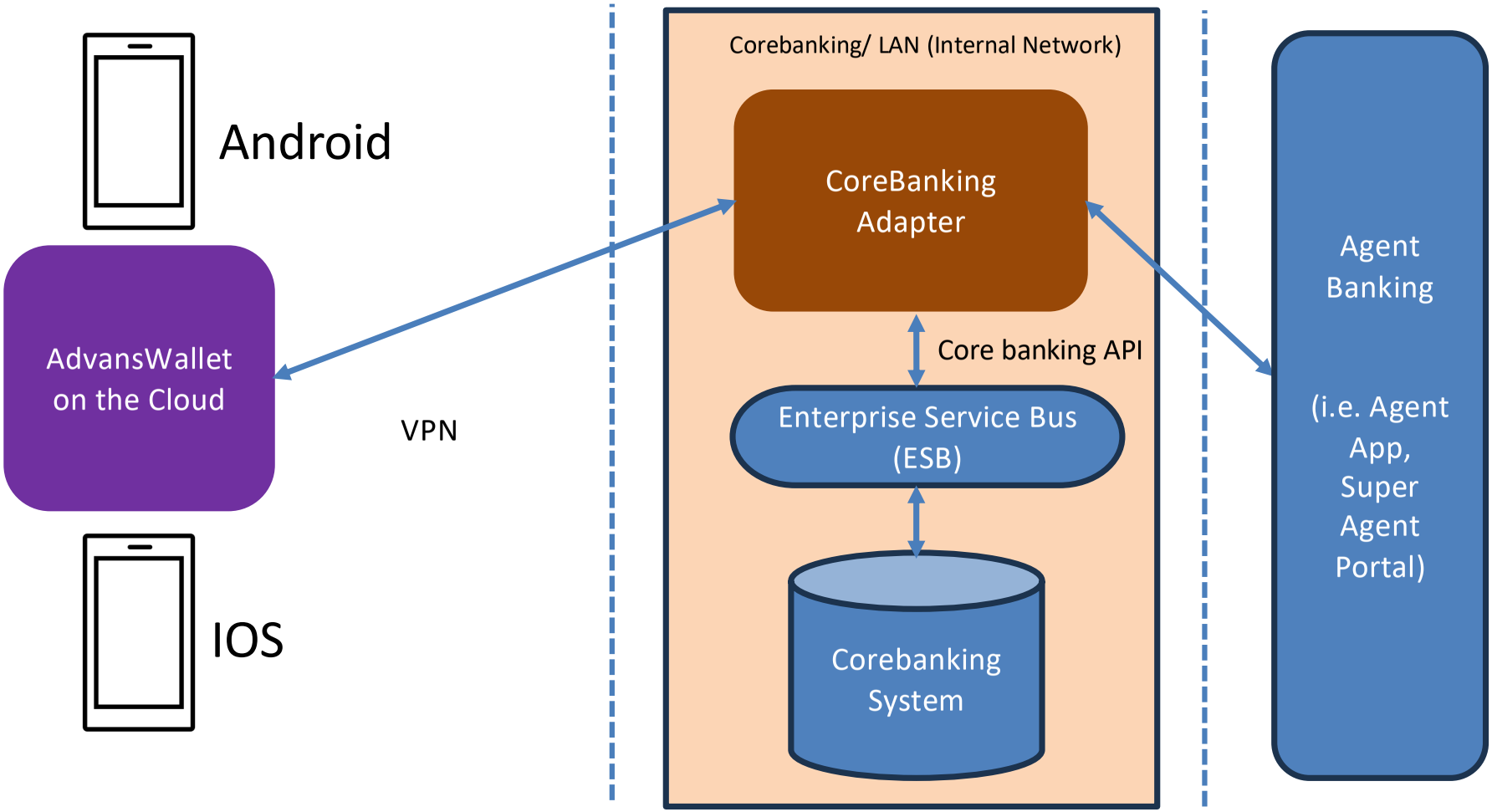
Service Architecture



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Security & Audit Module:

The solution incorporates 4 key security components

- **Role Management**

The system comes with user-defined role management and enforced through maker-checker approach.

- **Authentication and Authorization**

The system uses OAuth 2.0 which grants third-party applications only the data that they need.

- **Auditing and Logging**

Every user interaction on the system will be tracked and logged

- **Vulnerability Management and Penetration Testing**

The solution will be scanned on periodic basis to ensure there are no undetected vulnerabilities and security gaps.

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Technical Requirement:

In order to roll out this solution, we will need some resources from the Bank

Operating System	Linux (Ubuntu 22.04 or RHEL 7. x or CentOS 8.x) or Windows Server 2022
RAM	16GB
Storage	512GB
LANGUAGE AND PROGRAMMING FRAMEWORK	Java 21, JSON and Angular 16
Database Server	PostgreSQL or Oracle or MS SQL or MySQL
Container	Docker 4.24.2

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CoreBanking API Requirement – JSON!

1. Get Account Info

SN	Details	Value
1	Type	GET
2	Description	Get The account information of a single account
3	URL	<BASEURL>/AdvansWallet/getAcctInfo/1011010000910
4	Request	

5	Response	{
		"result": {
		"acctNum": "1011010000910",
		"acctTypeCode": "STDSAV",
		"acctTypeDesc": "STANDARD SAVINGS ACCOUNT",
		"acDesc": "OUMIE JAWARA",
		"acStatus": "DORMANT",
		"ccy": "GMD"
		},
		"errMsg": "Successful",
		"errCode": 0
		}

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CoreBanking API Requirement – JSON!

2. Get Account Balance Only

SN	Details	Value
1	Type	GET
2	Description	Get Account Balance Only
3	URL	<BASEURL>/AdvansWallet/getBalanceOnly/1011020004387
4	Request	

5	Response	{
		"avlBal": 4457206.497,
		"currBal": 4457206.497,
		"errCode": 0,
		"errMsg": "Successful"
		}

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CoreBanking API Requirement – JSON!

3. Process Transfer

SN	Details	Value
1	Type	POST
2	Description	Process transfer (account-Wallet/Wallet-Account
3	URL	<BASEURL>/AdvansWallet/processTransfer

4	Request	{
		"amt": 1,
		"desc": "testing",
		"frmAcct": "1011780031984",
		"refNo": "34243234232344",
		"toAcct": "1011740014883",
		“ccy”: ”GMD”
		}

5	Response	{
		"result": "0000000000000067",
		"errCode": 0,
		"errMsg": "SUCCESS"
		}

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CoreBanking API Requirement – JSON!

4. Get Transaction Status

SN	Details	Value
1	Type	GET
2	Description	Get Transaction Status
3	URL	<BASEURL>/AdvansWallet/ GetTransactionStatus/refNo
4	Request	

5	Response	{
		"result": null,
		"errCode": 0,
		"errMsg": "SUCCESS"
		}

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CoreBanking API Requirement – JSON!

5. Get Branches

SN	Details	Value
1	Type	GET
2	Description	Get Branches
3	URL	<BASEURL>/AdvansWallet/GetBranches
4	Request	

5	Response	{
		"result": [
		{
		“branchCode”:”001”,
		“branchName”:”Head Office”
		},
		{
		“branchCode”:”002”,
		“branchName”:”BO”
		}
		],
		"errCode": 0,
		"errMsg": "SUCCESS"
		}

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CoreBanking API Requirement – JSON!

6. LinkAccount

SN	Details	Value
1	Type	POST
2	Description	Link Account to Mobile APP
3	URL	<BASEURL>/AdvansWallet/LinkAccount

4	Request	{
		”AccountNo": 1011780031984,
		”MobileNo": ”220056789",
		”DateofBirth": ”19/03/1996",
		"refNo": ”43455456",
		}

5	Response	{
		"result": "00000000000000012",
		"errCode": 0,
		"errMsg": "SUCCESS"
		}

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CoreBanking API Requirement – JSON!

7. Get Loan Account Only

SN	Details	Value
1	Type	GET
2	Description	Get Loan Account
3	URL	<BASEURL>/AdvansWallet/GetLoanAccount

4	Request	{
		”customerID": 322421,
		}

5	Response	{
		”loanAccount": 32242132244,
		"errCode": 0,
		"errMsg": "Successful"
		}

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CoreBanking API Requirement – JSON!

8. Get Loan Repayment Amount

SN	Details	Value
1	Type	GET
2	Description	Get Loan Account Details
3	URL	<BASEURL>/AdvansWallet/GetLoanRepaymentAmount

4	Request	{
		“LoanAccount”: 32242132244,
		}

5	Response	{
		”repaymentAmount": 300.00,
		"errCode": 0,
		"errMsg": "Successful"
		}

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CoreBanking API Requirement – JSON!

9. Get Investment Account Only

SN	Details	Value
1	Type	GET
2	Description	Get Investment Account Details
3	URL	<BASEURL>/AdvansWallet/GetInvestmentAccount

4	Request	{
		”customerID": 322422,
		}

5	Response	{
		”investmentAccount": 32242212345,
		"errCode": 0,
		"errMsg": "Successful"
		}

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Contact us:

Email: info@oakwoodadvansio.com

Mobile: +233 244 179605

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